

Assignment No. 8

Group 16

Problem Statement

Perform detailed software cost estimation using Function Point Analysis (FPA) and Use Case Points (UCP) for the **Road Traffic Analysis and Prediction System**.

Abstract

Software cost estimation is a fundamental process in software engineering that enables developers and project managers to estimate the effort, cost, and time required for system development. Accurate estimation reduces project risks and improves resource planning.

This assignment applies two widely accepted estimation techniques: Function Point Analysis (FPA) and Use Case Points (UCP). These methods help quantify software size and complexity based on functionality and user interaction, specifically tailored for the Road Traffic Analysis and Prediction System.

Introduction

Software systems are becoming increasingly complex, making accurate estimation essential. Estimation techniques help determine the required effort, development time, and team size. The Road Traffic Analysis and Prediction System includes the following functionalities:

- User Registration and Authentication
- Real-time Traffic Prediction with Multi-factor Analysis
- Incident and Obstacle Reporting (Potholes, Construction, Diversions)
- Infrastructure and POI Management
- Historical Traffic Data Analytics and Visualization
- External API Integration (Google Maps, OpenWeather, Calendarific)

Function Point Analysis (FPA)

Function Point Analysis measures the functional size of a software system based on user interactions and data processing.

Step 1: Identify System Components

Component	Description	Count	Weight
External Inputs (EI)	User registration, login, complaint, events, paths.	6	4
External Outputs (EO)	Prediction results, analytics dashboards, exports.	4	5
External Inquiries (EQ)	Path search, POI search, log viewing.	4	4
Internal Logical Files (ILF)	User, Path, Prediction Log, Obstacles, POI, Complaints.	6	10
External Interface Files (EIF)	Google Maps, Weather, Calendarific, BMS Scraper.	4	7

Step 2: Calculate Unadjusted Function Points (UFP)

$$\begin{aligned}UFP &= (6 \times 4) + (4 \times 5) + (4 \times 4) + (6 \times 10) + (4 \times 7) \\&= 24 + 20 + 16 + 60 + 28 \\&= 148\end{aligned}$$

Step 3: General System Characteristics

There are 14 factors such as:

- Data Communications, Performance Requirements, Security Requirements, Transaction Rates, User Interfaces, Complex Processing.

Assume average rating = 3.35 (Total $\sum F_i = 47$):

$$\sum F_i = 47$$

Step 4: Value Adjustment Factor (VAF)

$$\begin{aligned}VAF &= 0.65 + (0.01 \times 47) \\&= 1.12\end{aligned}$$

Step 5: Adjusted Function Points

$$\begin{aligned}FP &= UFP \times VAF \\&= 148 \times 1.12 \\&= 165.76 \approx 166\end{aligned}$$

Step 6: Effort Estimation

Assume: 1 FP = 6 person-hours

$$Effort = 166 \times 6 = 996 \text{ hours}$$

Use Case Points (UCP)

Use Case Points estimate system complexity based on actors and use cases.

Step 1: Actor Classification

Actor Type	Description	Count	Weight
Simple	External APIs (Maps, Weather)	3	1
Average	System Scheduler	1	2
Complex	End users (UI based)	2	3

$$\begin{aligned}UAW &= (3 \times 1) + (1 \times 2) + (2 \times 3) \\&= 3 + 2 + 6 = 11\end{aligned}$$

Step 2: Use Case Classification

Use Case	Type	Count	Weight
User Auth & Profile	Simple	2	5
Incident Reporting	Average	3	10
Traffic Prediction	Complex	1	15
Data Analytics	Complex	1	15
Path Management	Average	2	10

$$\begin{aligned}UUCW &= (2 \times 5) + (3 \times 10) + (1 \times 15) + (1 \times 15) + (2 \times 10) \\&= 10 + 30 + 15 + 15 + 20 = 90\end{aligned}$$

Step 3: Unadjusted Use Case Points

$$UUCP = UAW + UUCW = 11 + 90 = 101$$

Step 4: Technical Complexity Factor (TCF)

Assume $TCF = 1.10$

Step 5: Environmental Factor (EF)

Assume $EF = 0.90$

Step 6: Final Use Case Points

$$\begin{aligned}UCP &= 101 \times 1.10 \times 0.90 \\&\approx 100\end{aligned}$$

Step 7: Effort Estimation

Assume: 1 UCP = 10 person-hours

$$Effort = 100 \times 10 = 1000 \text{ hours}$$

Comparison of Results

Method	Size	Effort (hours)	Approach
FPA	166 FP	996 hrs	Function-based
UCP	100 UCP	1000 hrs	Use-case based

Conclusion

Both methods produced comparable results, confirming that the system has high complexity due to its data-intensive nature and integration with various external services. These estimation techniques provide valuable insights for resource allocation, scheduling, and project management.